

Lluís Estapé Cusí

Audio & Speech Signal Processing · Real-Time DSP · Barcelona, Spain
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Project Portfolio (PDF)

RESEARCH INTERESTS

- Signal processing for speech and audio, and the engineering that has to sit underneath it before a model is useful in real time: voice activity detection, keyword spotting, feature extraction and the front-end pipelines that feed acoustic models. My working method is measurement-first: every project below ships an offline harness that quantifies what the system actually does, including where it does it worse, and those cases are reported rather than tuned around. My bachelor's thesis sits in a different corner of audio DSP (numerical antialiasing for nonlinear circuit models, advised at BME); what it has given me is the habit of deriving a method, benchmarking it against the prior art and validating it on hardware.

EDUCATION

- **Universitat Politècnica de Catalunya (UPC) — ETSETB / TelecomBCN** Barcelona, Spain
BSc Telecommunications Engineering — Audio & Video specialisation *Sep 2022 – Jun 2027 (expected)*
 - **Speech coursework:** Audio and Speech Processing (PAV) — implemented voice activity detection and speech-analysis pipelines in Python and C++, including framing, short-term energy and zero-crossing features, endpoint detection and the evaluation harness used to score them. Complemented by Audio and Video Signal Processing (PSAVC) and Acoustics & Electroacoustics.
 - **Acoustics project:** A 675-seat theatre modelled from scratch in EASE 4.4 and re-conditioned as a cinema: at 12.18 m³/seat, twice the usual target, the room was brought back to an Eyring RT of **0.8–1.2 s across 125 Hz–4 kHz**, then validated to UNE-EN ISO 3382-1 with mirror-image echograms, STI mapping under NC-25 and a dummy-head binaural auralisation. STI clears 0.6 in the stalls but falls to ≈ 0.55 on the rear balcony, a volume limit I traced and reported rather than tuned around; the cinema reconfiguration lifts the minimum to **0.64**.
- **Budapest University of Technology and Economics (BME)** Budapest, Hungary
Erasmus+ exchange — thesis research, Dept. of Artificial Intelligence & Systems Engineering *Spring 2027*
- **Frederic Mistral — Tècnic Eulàlia** Barcelona, Spain
Technological Baccalaureate *Sep 2020 – Jun 2022*

SPEECH & MACHINE LEARNING WORK

- **Spanish voice control for a real-time instrument (Aetheric Geometry):** Keyword spotting over a Vosk Spanish model running concurrently with a live audio callback and a computer-vision pipeline. The interesting part was not recognition but rejection: ordinary room sound fired **11 false commands per 40 s** before gating, so I measured the level and utterance-length distributions of true commands against background and built a triage stage that suppresses them without adding latency to the accepted path. The measurement script ships with the repository.
- **Voice activity detection and speech analysis (PAV, UPC):** Frame-level VAD in Python and C++ over short-term energy and zero-crossing-rate features, with endpoint detection and a scored evaluation harness against labelled references.
- **Neural models in applied projects:** GRU and transformer models (incl. FinBERT) for sequence and text tasks; MediaPipe and YOLOv8 pipelines in projects with real users. Comfortable with the PyTorch/NumPy side of an experiment and with keeping the data pipeline reproducible.

RESEARCH

- **Bachelor's Thesis — advised by Dr. Balázs Bank (BME)** Barcelona & Budapest
Antialiasing for nonlinear state-space virtual-analog models *Sep 2026 – Jun 2027*
 - **Contribution:** Antiderivative antialiasing (ADAA) has been folded into the trapezoidal rule (Holters, DAFx-19), into wave digital filters (Albertini et al., DAFx-20) and into the K-method; it has not been folded into the linearly-implicit Rosenbrock–Wanner solvers that DAFx-25 established for virtual analog. That is the gap I work in: re-deriving the half-sample delay compensation and frequency scaling for a multi-stage solver, and testing whether ADAA preserves its L-stability.
 - **Method:** A MATLAB bench sweeping ADAA order $\times$ solver $\times$ oversampling on diode-clipper and Tube Screamer circuits, then a real-time C++ implementation measured against the physical circuit on a board I fabricate, the hardware validation absent from the MATLAB-only prior work. Targeting a DAFx submission.

OPEN-SOURCE CONTRIBUTIONS

- **Surge XT — surge-synthesizer/surge:** Contributor to a widely used open-source synthesizer; two pull requests, one merged. PR #8543 fixes an oscillator onset defect (issue #7570): the three BLIT oscillators seeded remaining *phase space* rather than *phase*, so with retrigger off every voice started late, extbf665.6 oversampled samples at MIDI 24 over 300 initialisations. Maintainer review found three defects I had missed, among them a period seeded without sync that ran the first block at extbf48% of the correct rate; each fix shipped with a before-and-after measurement. I also collapsed three duplicated copies of the oscillator state machine into one, and for the new tests broke the oscillator deliberately to check they could fail: my first design passed on broken code, so I replaced it and documented the failure mode the final three still miss.

- **OmniSim** — [omnilink-tech/omnisim](#): Four accepted contributions to a Webots-derived robotics simulator on Newton/MuJoCo physics, credited by name in the v8.4.0 changelog. A drone navigation regression I wrote exposed a defect no existing test could see: clearance was scored from the body origin instead of the oriented collider, so every flight reported 8/8 waypoint arrivals while the airframe cleared the shelving by millimetres. Once the maintainers shipped the fix, completion went from extbf8/18 to 18/18 over eighteen flights. My clearance figures also contradicted their published regression threshold by a factor of two, which they re-measured and withdrew. Later work turns the analysis into a tested scoring module, retracts an inference of my own that re-scoring refuted, and adds a documented force and torque helper whose test asserts the delivered impulse against the analytic value.

SELECTED AUDIO PROJECTS

FULL PORTFOLIO REPORT (PDF)

- **ESP** — **audio plugin suite (C++ / JUCE 8)**: A self-published brand of real-time VST3 instruments and effects. SYNTH/1's oscillator is a mip-mapped bank: 4 waveforms $\times$ 21 half-octave bands, 4-point Hermite interpolation, double-precision phase. An offline harness (`measure_oscillator.cpp`, compiled against a minimal JUCE stub) puts sawtooth alias rejection **96 dB better at A4 and 139 dB better at C8** than the previous linear-interpolated version, and 4 dB *worse* below 86 Hz, a table-image artefact I characterised rather than hid. Audio $\leftrightarrow$ UI transport is lock-free (atomics, ring buffers) with an allocation-free render callback. [esp-plugin-store.vercel.app](#)
- **AirStems** — **real-time gestural stem remixer: LALAL.AI Special Prize**, global Musixmatch Musicathon 2026, followed by a published interview and an invited technical guest article. Stem gains are quantised to a librosa-derived beat grid with click-free ramps inside a `sounddevice` callback (44.1 kHz, 512-sample blocks, audio thread measured at **$\sim 11\%$ load** with computer vision running concurrently), driven by two-handed MediaPipe tracking with hysteresis debouncing. Integrates LALAL.AI source separation, word-level synced Musixmatch lyrics and Cyanite analysis. [github.com/lluiseurope-upc/AirStems](#)
- **Aetheric Geometry** — **gestural additive-synthesis instrument**: The instrument the voice control above is built into: a six-gesture MediaPipe vocabulary resolved through a debounced state machine, driving additive voices on exact just-intonation ratios with hand-mapped TPT filtering and a Schroeder reverb. Every DSP claim is reproducible via `analysis/measure_dsp.py`. [github.com/lluiseurope-upc/AethericGeometry](#)
- **T-8 Groovebox** — **16-step sequencer, PCB design (KiCad)**: A 290×210 mm two-layer board taken from schematic to fabrication package: 98 components, 2,574 track segments, 1,150 vias, a ground pour on both layers tied by 417 stitching vias, **DRC clean with zero unrouted nets**. An ESP32-WROOM-32 drives two MCP23017 expanders and one I²S bus feeds two slave DACs in parallel (PCM5102 line out, MAX98357A class-D speaker). The audit caught two defects invisible to ERC, including an amplifier symbol on a mismatched footprint that left the audio outputs with no pads at all.
- **PAE-EDISA** — **autonomous inventory drone (UPC $\times$ EDISA)**: Ground Control Station lead (WP2) in a 14-student team delivering to a real client: a cross-platform Electron GCS (telemetry, 3D point-cloud viewer, SQLite inventory) over a rosbridge link to a ROS2 stack running Point-LIO LiDAR SLAM at ~ 800 k points per scan.

EXPERIENCE

- **Deloitte** Barcelona, Spain
Cybersecurity Strategy Consultant — risk assessments for regulated-sector clients *Feb 2026 – Jun 2026*
- **NTT Docomo** — **Mobile World Congress** Barcelona, Spain
Exhibition staff — demoed NTT QONOQ products to an international audience *Mar 2025*
- **Mundo Cero / Arcadia Motion Pictures** Bilbao & Barcelona, Spain
Production assistant — feature films “Como salvar el mundo” and “Un novio para mi mujer” *2021, 2025*

TECHNICAL SKILLS

- **Programming**: Python, C++ (17/20), C, MATLAB, JavaScript (React, Electron), Java
- **Speech & ML**: voice activity detection, keyword spotting, Vosk, short-term spectral and energy features, evaluation harnesses; GRU and transformer models, PyTorch, NumPy, librosa, `sounddevice`, MediaPipe, OpenCV, YOLOv8
- **Audio DSP**: IIR/FIR and TPT filters, FFT analysis, oversampling & antialiasing, band-limited oscillators, algorithmic reverb, dynamics processing, lock-free real-time architectures, JUCE 8 / VST3; room acoustics: EASE 4.4 modelling, RT prediction, STI mapping, binaural auralisation
- **Tools & practice**: Git, CMake, pytest, Linux, LaTeX; offline measurement harnesses as a standard part of every change
- **Languages**: Catalan & Spanish (native), English (Cambridge Advanced, C1), French (basic)